MACHINE LEARNING

HOMEWORK – 3

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Problem – 1

We've avoided explicitly forming an R^p Bag of Words vector by realizing that for any given review its associated vector will be mostly 0 because no review will contain anywhere near the total words possible, and so by making this realization we decided to build a dictionary with the numerical I.D.'s of each word identified in a particular review mapped to a value that is the number of times it appears in the written review.

Henceforth, we defined a function for computing the dot product without having to parse through the entire weight vector by taking in the aforementioned dictionary we just described and then finding the index in the weight vector associated with every particular key inside the dictionary and then summing the products of the values at each key in the vector and the values at those indices in the weight vector. This way it explicitly avoids needing to parse 2 R^p vectors to take the dot product between them.

This algorithm with online perception provides error rates of 0.178 on the training data set and 0.179 on the test data set.

Code Output:

```
TRAINING ERROR RATE: 0.178736
done converting

TEST ERROR RATE: 0.1793441250523238
```

Problem – 2

The top 10 most positively weighted words are: 'perfection', 'disappoint', 'gem', 'phenomenal', 'perfectly', 'heavenly', 'fantastic', 'perfect', 'awesome', 'skeptical' The top 10 most negatively weighted words are: 'worst', 'mediocre', 'meh', 'underwhelmed', 'worse', 'bland', 'tasteless', 'flavorless', 'hopes', 'lacked'

Code Output:

```
 TOP 10 GOOD WORDS:
 ['perfection', 'disappoint', 'gem', 'phenomenal', 'perfectly', 'heaven
ly', 'fantastic', 'perfect', 'awesome', 'skeptical']

 TOP 10 BAD WORDS
 ['worst', 'mediocre', 'meh', 'underwhelmed', 'worse', 'bland', 'tastel
ess', 'flavorless', 'hopes', 'lacked']
```

Problem – 4

All the python codes are in the zip folder.

A.

```
number of features in j_hat after step-2:  42
```

B.

```
The emperical risk of w_hat after step 3:  0.213
We see that this value is much smaller than 1, which is the closeby estimate of the variance of the labels on this dataset
```

C.

```
Risk of w_hat on the test set:  1.403
```

D.

We know that empirical risk is computed over the training samples (dataset), and the true risk is estimated over the entire population from which the samples have been derived.

Mathematically, we can see that:

$$Risk[\widehat{w}] = E\left[loss_{sq}(\widehat{w}^T X, Y)\right]$$

Given the small number of features and data points used for estimation of $\widehat{w}$, it may seem as though the features and the labels are correlated, however, they are not as all the features and the labels have been drawn independently from a standard normal distribution. This happens because the Freedman's paradox is an extreme case of model selection bias. The model selected according to Freedman's correlation threshold criteria somehow provides a very low value of empirical risk, leading to the belief that the features and labels are correlated, when they are actually not.

E.

I feel that the empirical risk on the second dataset should be higher than the empirical risk on the first dataset. This is because $\hat{J}$ has been computed over the first dataset, and thus assumes a correlation between the features and the labels of the first dataset. However, since the second dataset (which is independent of the first), $\hat{J}$ fails to take into the account the correlation between the features and the labels of the second dataset.

Since, the two datasets are independent to each other, there is no logical way to know whether the same features are correlated to the same labels in the second dataset as the first dataset.

Thus, when we compute $\widehat{w}$ over the second dataset using $\hat{J}$ obtained from the first dataset, we should get an empirical risk that is higher than the empirical risk obtained from computing $\widehat{w}$ over the first dataset.

The following results have been obtained from python code:

```
The emperical risk of w_hat computed from second dataset:  1.926
```

Thus, we observe that the empirical risk of $\widehat{w}$ for the second dataset is much higher than the empirical risk of $\widehat{w}$ for the first dataset (which was 0.213).

Problem – 5

A.

By definition,

NS(A) is the set of all vectors $\bar{X}$, such that $A.\bar{X} = 0$.

$$NS(A) = \{\bar{X}, such\ that\ A.\bar{X} = 0\}$$

$$NS(A^T.A) = \{\bar{X}, such\ that\ A^T.A.\bar{X} = 0\}$$

Hence, we can clearly see that:

NS(A) contains $\bar{X}$, such that,

$$A.\bar{X} = 0 \qquad (1)$$

We know,

$$A^T.A.\bar{X} = A^T.(A.\overline{X})$$

$$A^T.A.\bar{X} = A^T.(A.\overline{X})$$

$$A^T.A.\bar{X} = A^T.0$$

$$A^T.A.\bar{X} = 0$$

Therefore, for every vector $\bar{X}$ in the $NS(A)$, the same vector $\bar{X}$ exists in $NS(A^T.A)$ as well. Therefore, we may conclude that the two sets $NS(A), and\ NS(A^T.A)$ are equal sets.

Since, $NS(A^T.A)$ and $NS(A)$ are equal sets,

$$NS(A) = NS(A^T.A)$$

B.

We are given an $n\ X\ d$ matrix A.

By definition,

We know that,

$$CS(A) = \{\ b \in R^n \mid b = Ax, x \in R^d\ \} \tag{1}$$

$$CS(AA^T) = \{\ b \in R^n \mid b = (AA^T)x, x \in R^n\ \} \tag{2}$$

Considering (1),

$$CS(A) = \{\ b \in R^n \mid b = Ax, x \in R^d\ \}$$

We know from principle of rank nullity that, any vector x shall be decomposed as:

$$x = u + v, such\ that\ u \in NS(A)\ and\ v \in CS(A^T)$$

$$b = A(u + v)$$

$$b = Au + Av$$

$As, u \in NS(A),$

$Au = 0$

$Therefore,$

$b = Av$

$As, we\ know\ that, v \in CS(A^T)$

$v = A^T w, \quad w \in R^n$

$\Rightarrow\ b = AA^T w, \quad w \in R^n$

$\Rightarrow b = (AA^T)w, \quad w \in R^n$

$Thus, from\ (2),$

$$CS(A) \subseteq CS(AA^T) \tag{3}$$

Considering (2),

$CS(AA^T) = \{ b \in R^n \mid b = (AA^T)x, x \in R^n \}$

$b = AA^T x$

$b = A(A^T x)$

$Let\ A^T x = w, \qquad w \in R^d$

$\Rightarrow b = Aw$

$\Rightarrow b \in CS(A)$

$Thus, from\ (1),$

$$CS(AA^T) \subseteq CS(A) \tag{4}$$

$from\ (3)\ and\ (4),$

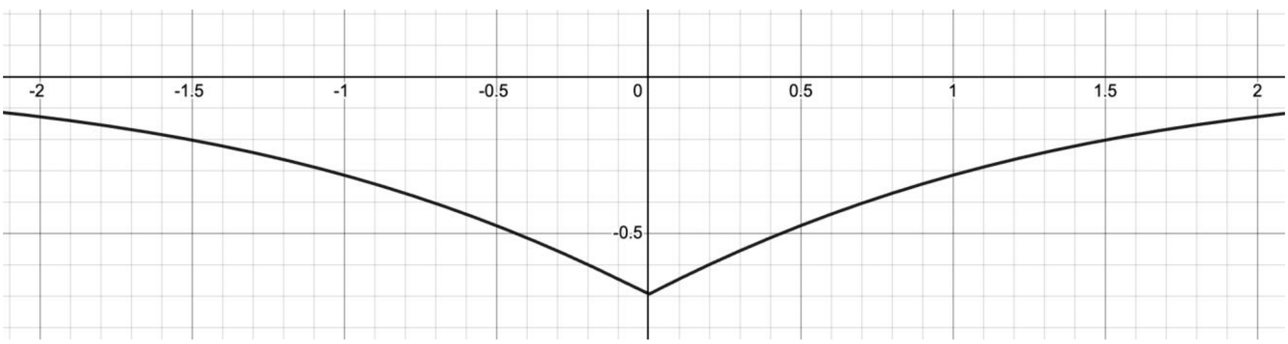
$CS(AA^T) \subseteq CS(A)\ and\ CS(A) \subseteq CS(AA^T)$

$\Rightarrow CS(A) = CS(AA^T)$

Problem – 6

γ , $\hat{w}$, LL(w), and the condition that y=1 if and only if, $\hat{w}^T x > 0$ can be used to get a lower bound on what LL(w) could have been the condition between y and $\hat{w}^T x$ means that when the summand, if we view $\hat{w}^T x$ as one argument, is a symmetric function always less than 0 and monotonically increase in the absolute value of its argument. this is because:

$$1 * x - ln(1 + e^x) = -ln(1 + e^{-x})$$



So, we can use this quality to get a lower bound on what $\mathrm{LL}(\hat{\mathrm{w}})$ could have been which is:

$\mathrm{LL}(\hat{\mathrm{w}})_{\min} = \mathrm{n}\left(\gamma - \ln\left(1 + \mathrm{e}^{\gamma}\right)\right)$

because if you assume that every $\mathrm{x_i}$ in the input set was the vector giving the lowest $\hat{\mathrm{w}}^{\mathrm{T}}\mathrm{x}$ which gives the lowest summand which then gives the lowest possible sum in the LL(w) function which would be the most negative value that LL(w) could take and therefore is the lower bound.

And so, this number $\mathrm{LL}(\hat{\mathrm{w}})_{\min}$ is a lower bound on what the log likelihood of $\hat{\mathrm{w}}$ could have been.

And so if use this number as starting point we can take the derivative of the LL(w) with respect to w which will give use the direction us steepest ascent and so if we us a steepest ascent iteration method where we start v at $\hat{\mathrm{w}}$ and then use the derivative of the LL(v) function to guide use toward greater values of LL(v) then this method will give use LL(v) where v with necessarily be greater than some negative number -ϵ

Problem – 7

$Considering\ a\ feature\ map, \varphi: R^d \to R,$

$such\ that, \varphi(\mathrm{x}) = (\mathrm{r} - \ || \ x\ -\ c\ ||)^3; r > 0, x \in R^d$

$Hence, for\ this\ feature\ map, p\ =\ 1$

$CASE - 1:$

$Let\ f_{c,r} = 1,$

$\Rightarrow\ ||\ x\ -\ c\ ||\ <\ r$

$\Rightarrow r\ -\ ||\ x\ -\ c\ ||\ >\ 0$

$\Rightarrow\ \varphi(\mathrm{x})\ =\ (r\ -\ ||\ x\ -\ c\ ||)^3$

$\Rightarrow \varphi(\mathrm{x})^T w\ =\ w.(r\ -\ ||\ x\ -\ c\ ||)^3;\ \ w \in R^d$

$So, we\ can\ see\ that\ for\ w\ =\ 1,$

$\varphi(\mathrm{x})^T w > 0\ \ \ \forall\, x \in R^d$

$CASE - 2:$

$Let\ f_{c,r} = 0,$

$\Rightarrow \; || x - c || \geq r$

$\Rightarrow r - || x - c || \leq 0$

$\Rightarrow \; \varphi(\mathrm{x}) = (r - || x - c ||)^3$

$\Rightarrow \varphi(\mathrm{x})^T w = w.(r - || x - c ||)^3; \; w \in R^d$

$So, we\ can\ see\ that\ for\ w = 1,$

$\varphi(\mathrm{x})^T w < 0 \quad \forall\, x \in R^d$

$Thus, the\ feature\ map\ \varphi\ is\ such\ that,$

$\exists w \in R, i.e. w = 1, such\ that\ \varphi(\mathrm{x})^T w > 0,$ if and only if $f_{c,r} = 1 \; \forall \; x \in R^d$

Hence, the feature map provided is such that homogeneous linear classifiers in the feature space are able to represent all binary classifiers of the given form.